

1. What is statically typed and dynamically typed Programming Language?

Statically typed languages require variables to be declared with their data types at compile time, and the type cannot change during execution. Java is statically typed - you must declare a variable's type when you create it.

Dynamically typed languages determine variable types at runtime, and a variable's type can change. Examples include Python and JavaScript.

2. What is the variable in Java?

A variable in Java is a container that holds data values during program execution. It has:

- A name (identifier)
- A data type
- A value (which can change, unless it's declared as final)

Example: `int age = 25;`

3. How To Assign a Value To Variable?

You assign values to variables using the assignment operator `=`:

java

Copy

Download

```
// Declaration and assignment in one statement
```

```
dataType variableName = value;
```

```
// Or separately:
```

```
dataType variableName; // declaration
```

```
variableName = value; // assignment
```

Example:

java

Copy

Download

```
int count;    // declaration
```

```
count = 10;   // assignment
```

```
double pi = 3.14; // declaration and assignment
```

4. What are Primitive Data types in Java?

Java has 8 primitive data types:

1. byte - 8-bit integer (-128 to 127)
2. short - 16-bit integer (-32,768 to 32,767)
3. int - 32-bit integer (-2^{31} to $2^{31}-1$)
4. long - 64-bit integer (-2^{63} to $2^{63}-1$)
5. float - 32-bit floating point
6. double - 64-bit floating point
7. char - 16-bit Unicode character
8. boolean - true or false

5. What are the Identifiers in Java?

Identifiers are names given to variables, methods, classes, etc. Rules:

- Can include letters, digits, underscore (`_`), dollar sign (`$`)
- Cannot start with a digit
- Cannot be a Java keyword
- Are case-sensitive

Examples: age, userName, MAX_VALUE, \$salary

6. List the Operators in Java

Java operators include:

1. Arithmetic: `+`, `-`, `*`, `/`, `%`
2. Assignment: `=`, `+=`, `-=`, `*=`, `/=`, `%=`
3. Comparison: `==`, `!=`, `>`, `<`, `>=`, `<=`
4. Logical: `&&`, `||`, `!`
5. Bitwise: `&`, `|`, `^`, `~`, `<<`, `>>`, `>>>`
6. Unary: `+`, `-`, `++`, `--`, `!`
7. Ternary: `? :`
8. Instanceof: `instanceof`

7. Explain about Increment and Decrement operators and give examples

Increment (`++`) and decrement (`--`) operators change a variable's value by 1.

Prefix form (`++x`, `--x`): Operation is performed first, then the value is used

Postfix form (`x++`, `x--`): Value is used first, then the operation is performed

Examples:

java

Copy

Download

```
int x = 5;
```

```
int y = x++; // y = 5, x = 6 (postfix)
```

```
int a = 5;
```

```
int b = ++a; // b = 6, a = 6 (prefix)
```

```
int count = 10;
```

```
count--; // count becomes 9
```

```
--count; // count becomes 8
```